

# Anomaly Detection & Dimensionality Reduction

Uncover hidden anomalies using cutting-edge ML techniques on real-world datasets.

## 1 Project Overview

I applied t-SNE, UMAP, and Isolation Forest for anomaly detection in two datasets. These methods effectively identified rare anomalies in jet engine and silicon thermal sensor data. Key results include successfully isolating anomalies and visualizing complex patterns.

**t-SNE:** Reduces dimensions while preserving local relationships.

**UMAP:** Offers faster dimensionality reduction while maintaining global structure.

**Isolation Forest:** Detects anomalies by randomly isolating outliers.

## 2 Datasets

### 2.1 Jet Engine Sensors

| Property           | Value                                                     |
|--------------------|-----------------------------------------------------------|
| Description        | Data from jet engine sensor readings with rare anomalies. |
| Total Rows         | 4,650                                                     |
| Columns            | 8                                                         |
| Features           | Turbine temperature, Oil pressure, Fan speed, etc.        |
| Class Balance      | 3.2% anomalies                                            |
| Train / Test Split | 80/20                                                     |

### 2.2 Silicon Thermal Hotspots

| Property           | Value                                              |
|--------------------|----------------------------------------------------|
| Description        | Thermal and electrical metrics from silicon chips. |
| Total Rows         | 9,300                                              |
| Columns            | 8                                                  |
| Features           | Thermal voltage, Leakage current, IR drop, etc.    |
| Class Balance      | 3.2% anomalies                                     |
| Train / Test Split | 80/20                                              |

## 3 Methodology

### 3.1 Algorithms Used

| Algorithm        | Description                                                 |
|------------------|-------------------------------------------------------------|
| t-SNE            | Chosen for visualizing high-dimensional data.               |
| UMAP             | Selected for its speed and effectiveness in large datasets. |
| Isolation Forest | Ideal for detecting anomalies in unlabeled data.            |

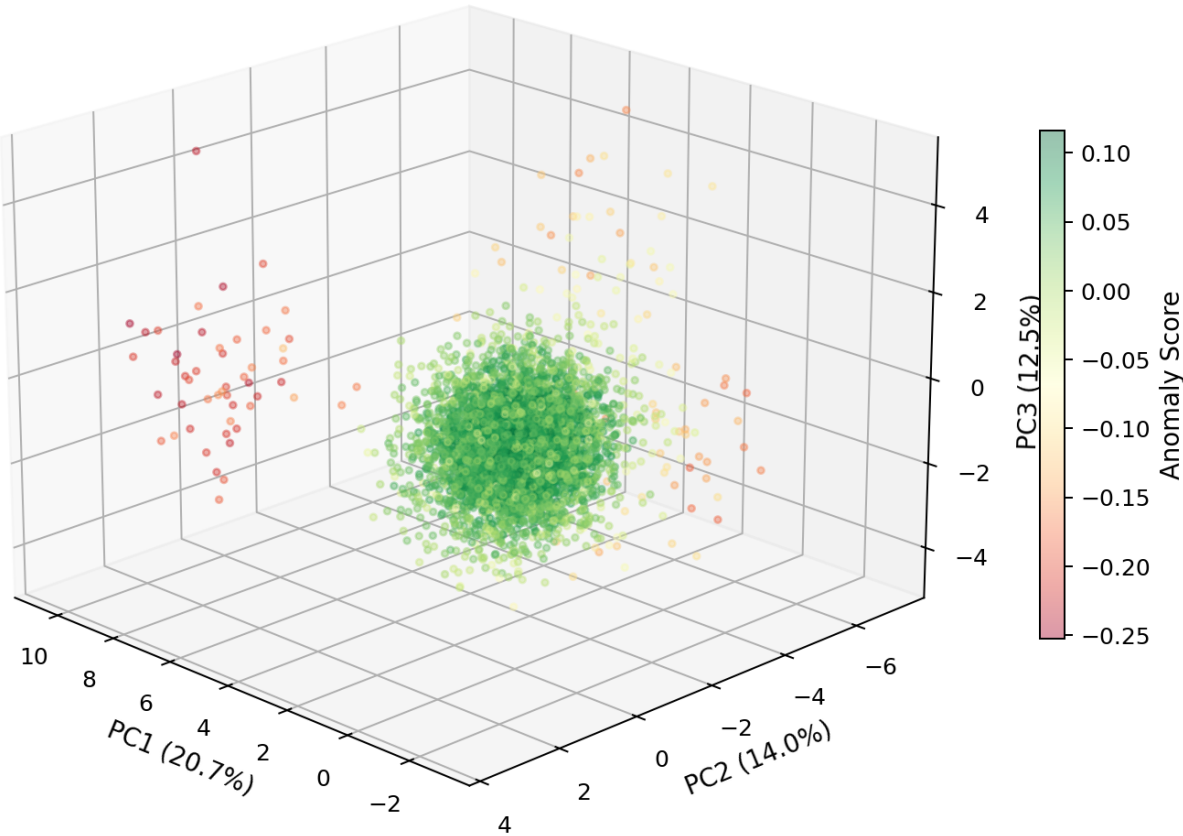
I loaded the datasets, performed exploratory data analysis, applied Isolation Forest for anomaly detection, and used t-SNE and UMAP for effective visualization of results.

## 4 Results

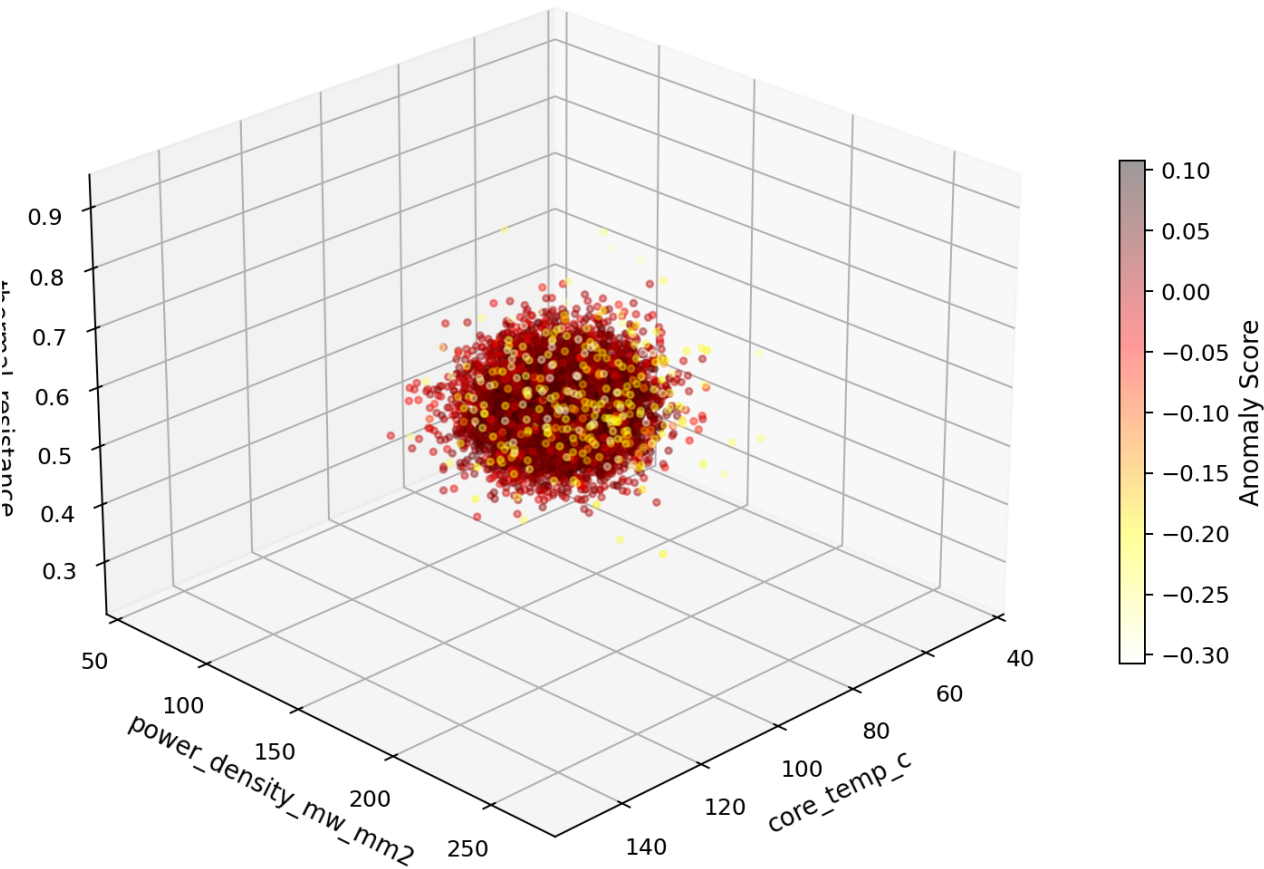
Isolation Forest emerged as the strongest model for anomaly detection due to its efficiency with high-dimensional data. I found that t-SNE and UMAP provided insightful visualizations, clearly highlighting underlying patterns and anomalies.

| Metric                         | Value                                                    |
|--------------------------------|----------------------------------------------------------|
| Isolation Forest Anomaly Score | Higher scores for isolated anomalies                     |
| Best Result                    | Anomaly score distribution shows well-defined isolation. |

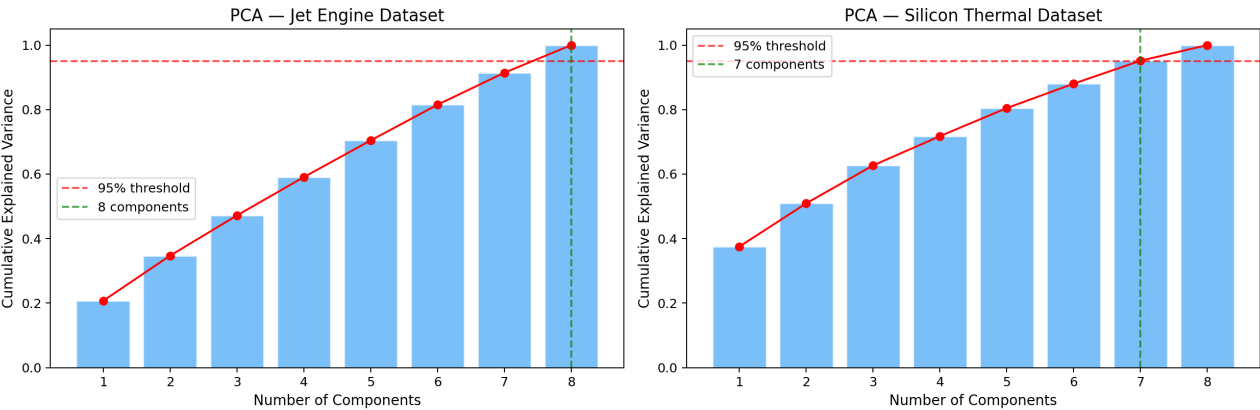
Isolation Forest Anomaly Scores in PCA Space  
Jet Engine Sensor Data

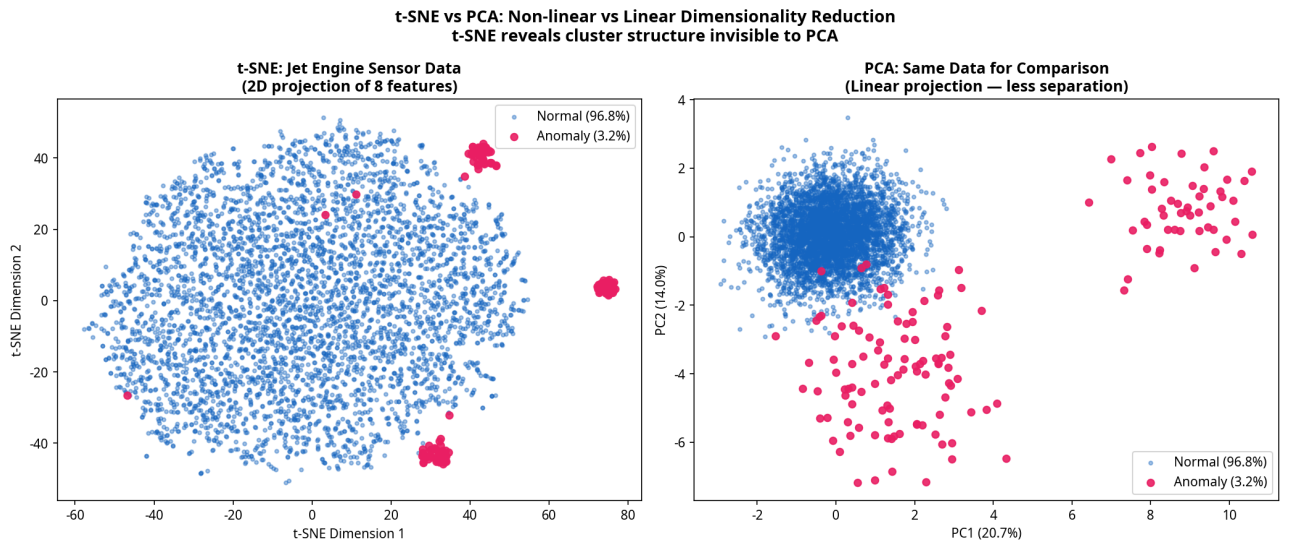
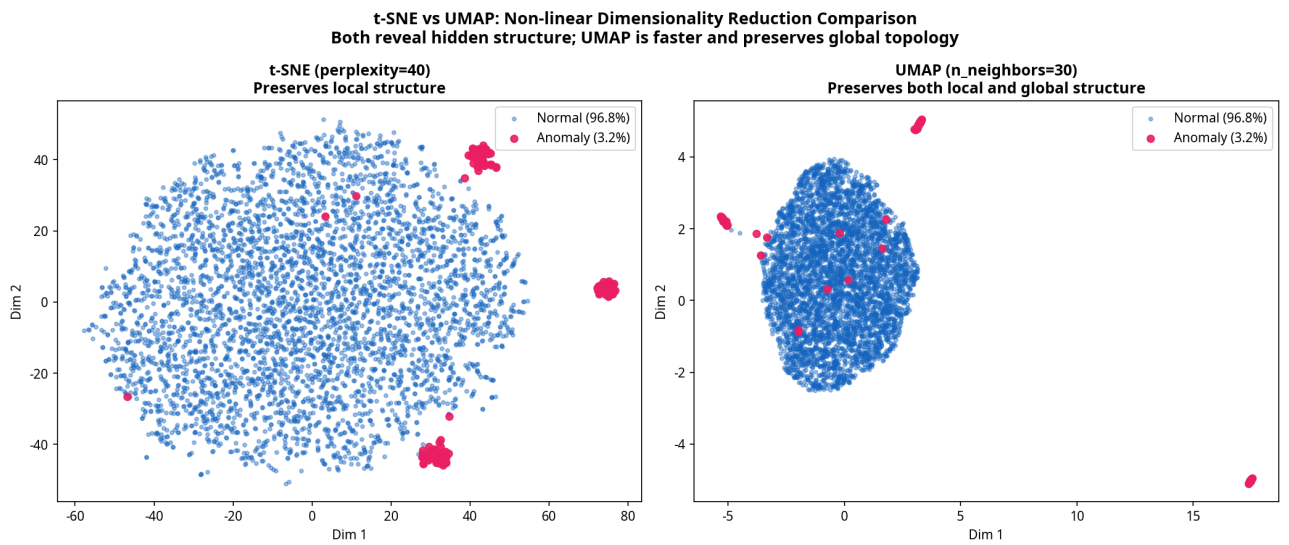
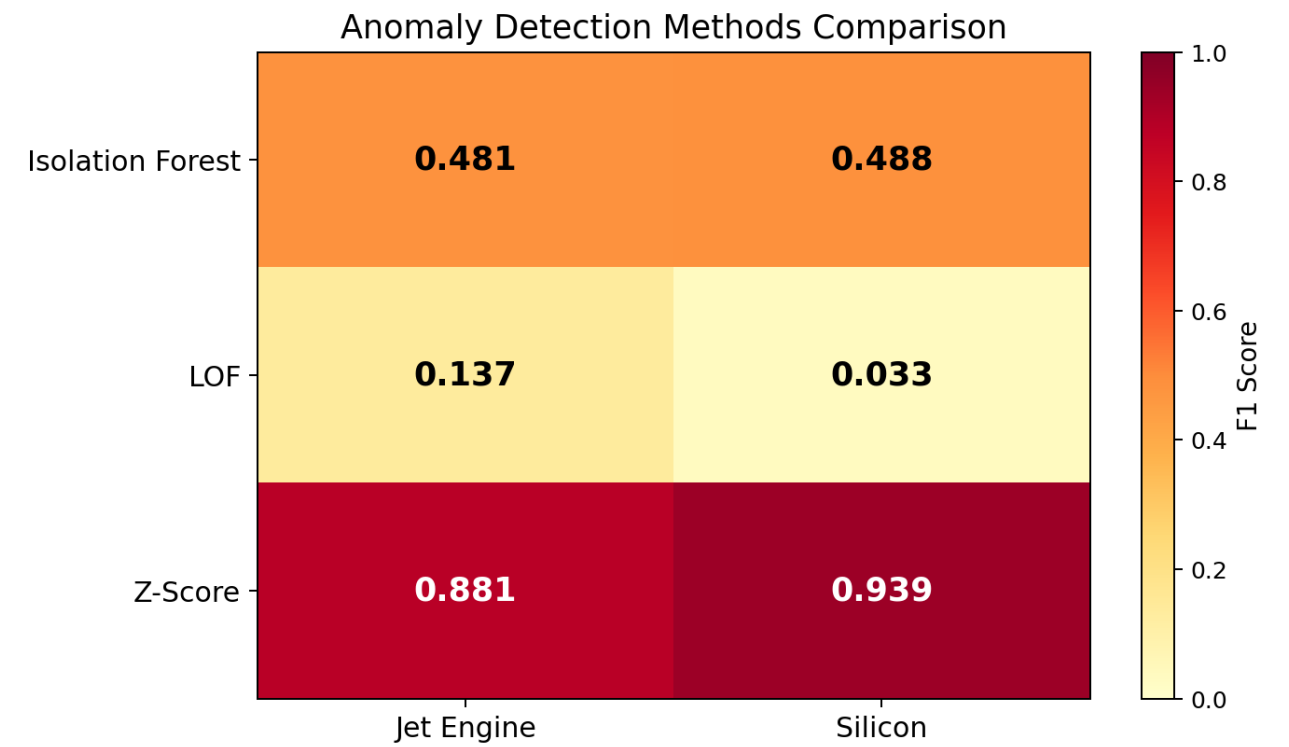


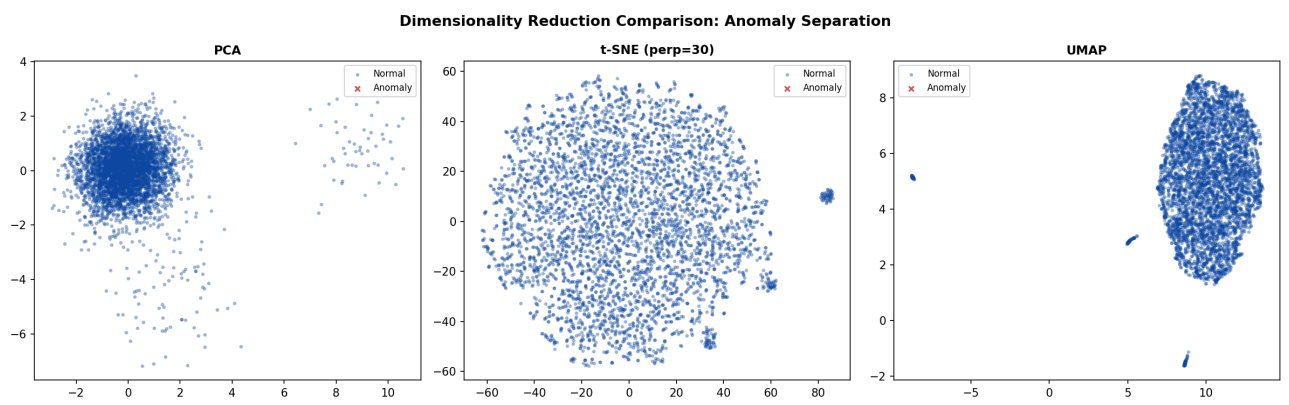
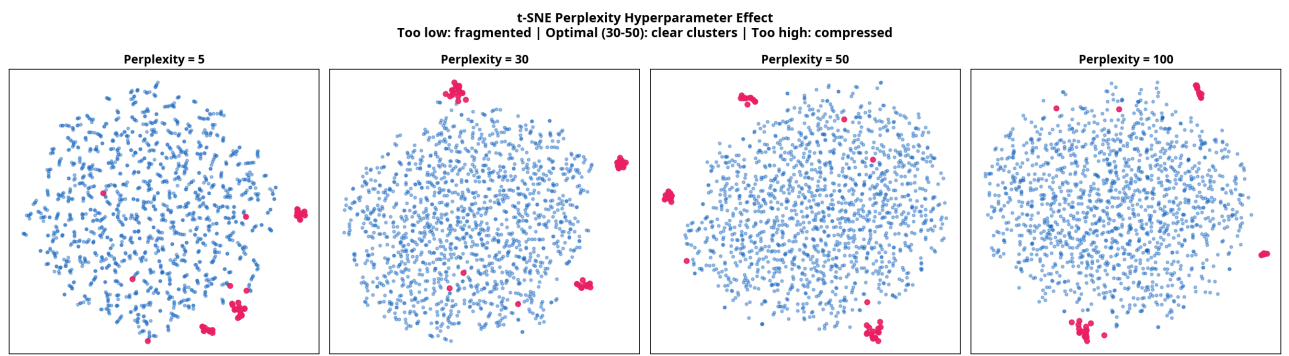
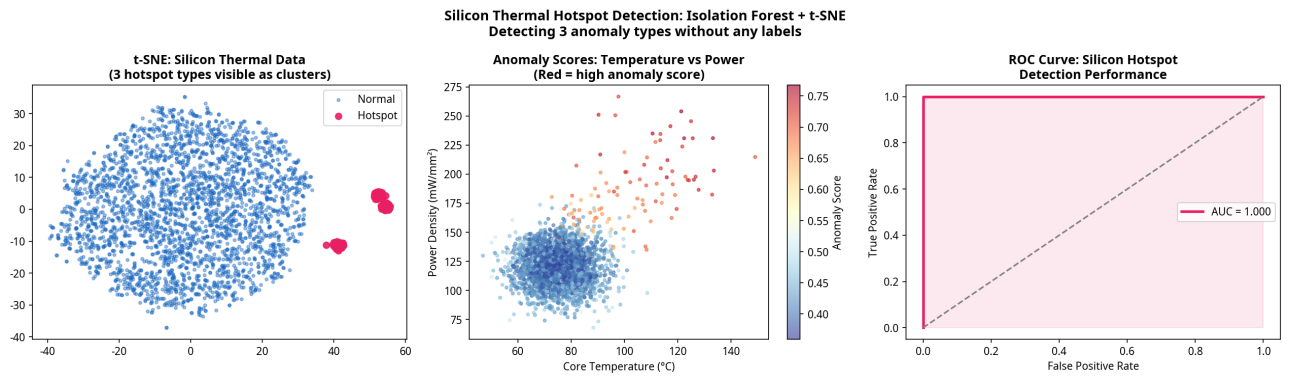
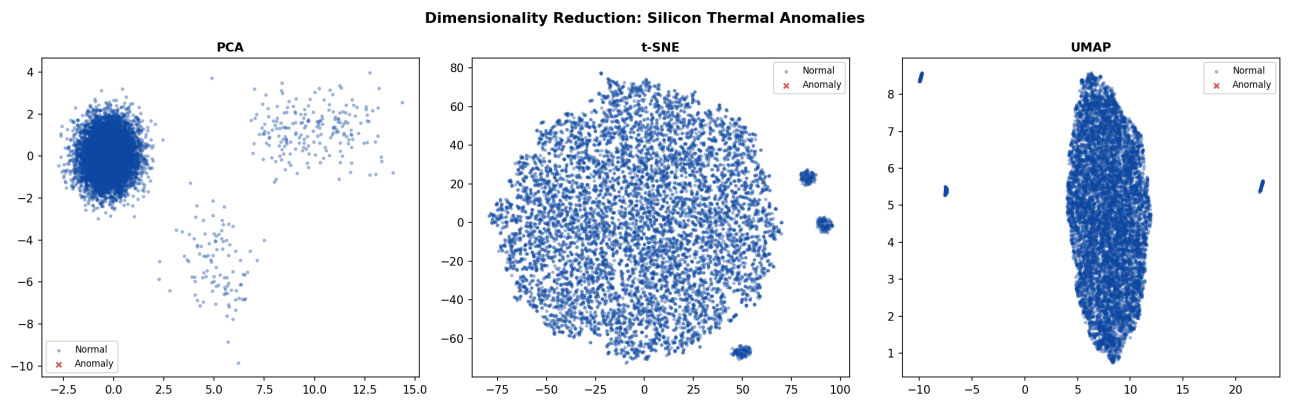
### Silicon Thermal Hotspot Detection Isolation Forest Scoring

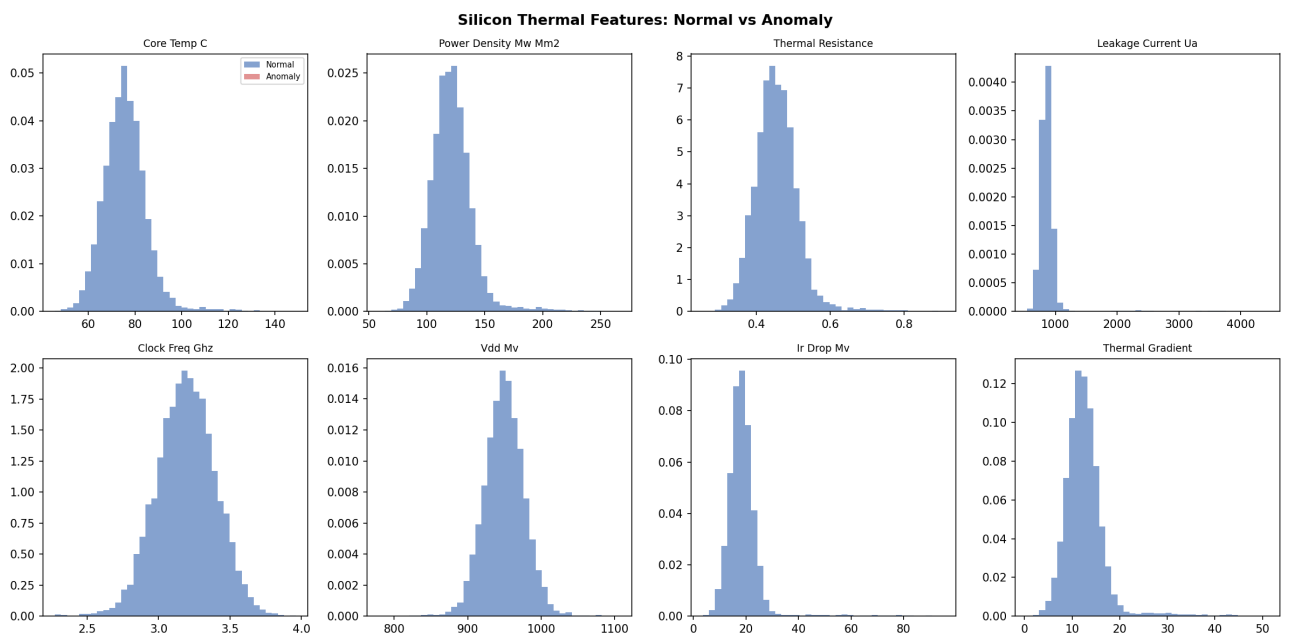
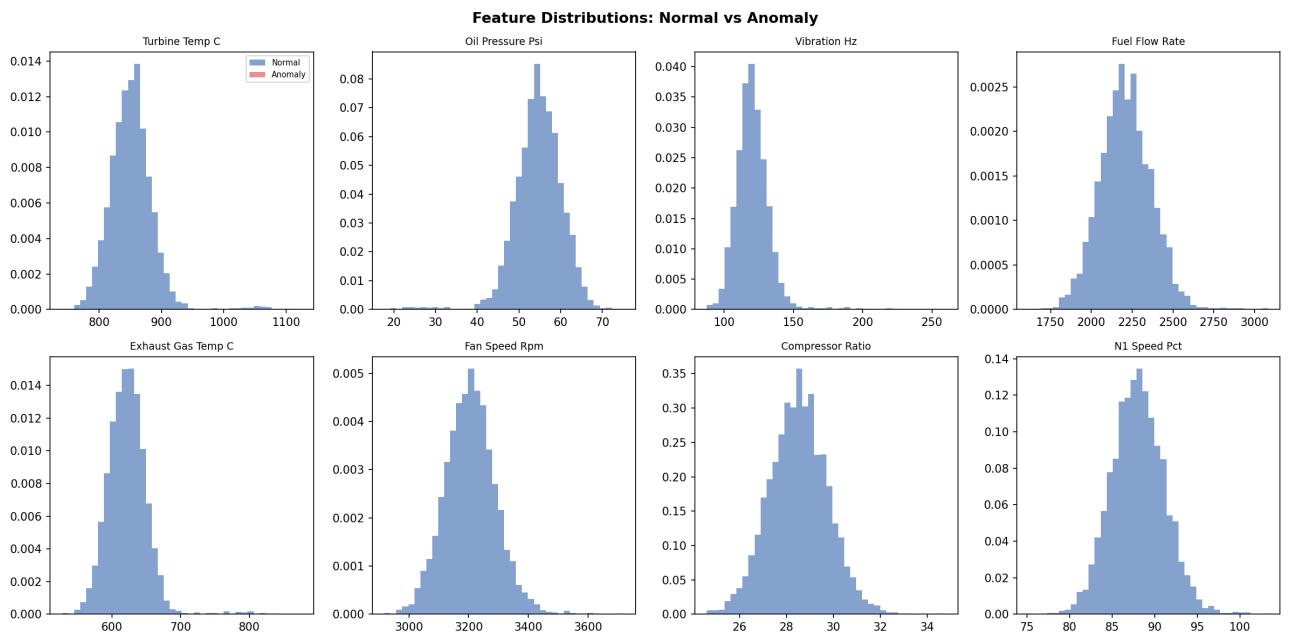
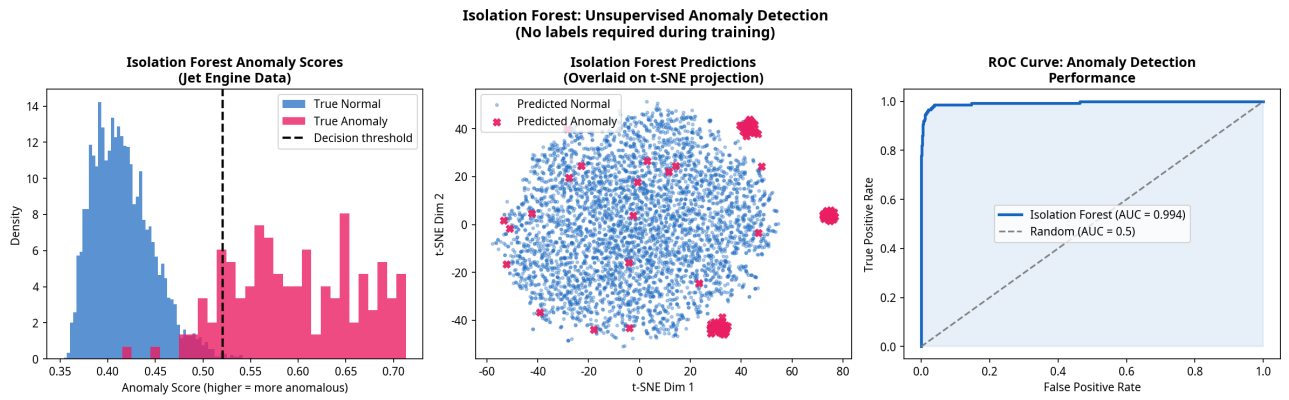


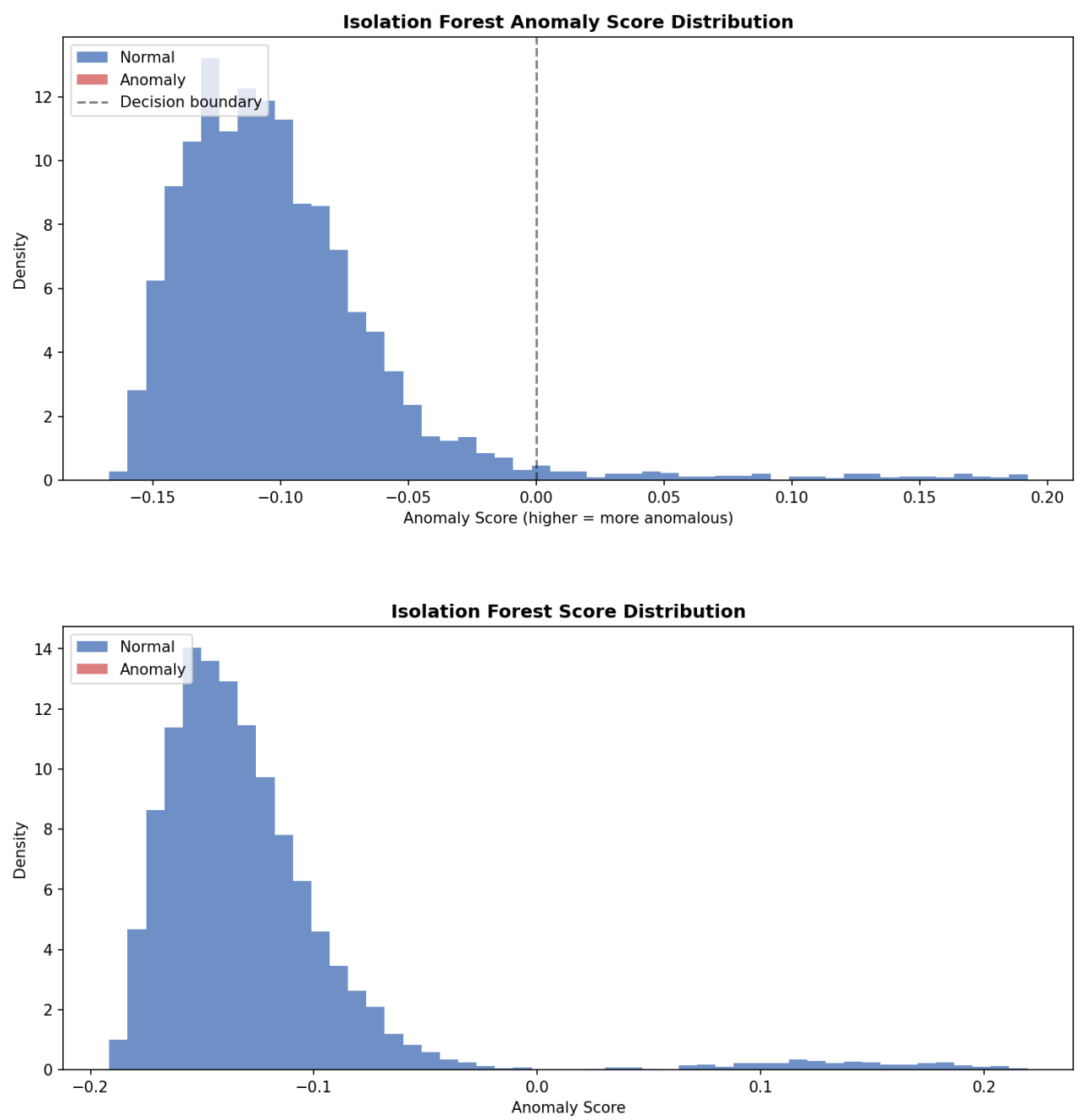
Dimensionality Reduction Analysis











Visualizations illustrate the distribution of anomalies and the effectiveness of dimensionality reduction techniques.

## 5 Tech Stack & Key Learnings

| Info       | Detail                                                                                                                                                                  |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub     | <a href="https://github.com/AIML-Engineering-Lab/007_anomaly_detection_dimensionality">https://github.com/AIML-Engineering-Lab/007_anomaly_detection_dimensionality</a> |
| Tech Stack | Python 3.11, scikit-learn, pandas, Matplotlib                                                                                                                           |
| Licence    | MIT                                                                                                                                                                     |

- Dimensionality reduction enhances anomaly detection.
- Isolation Forest effectively identifies rare events.
- Visualizations greatly aid in understanding complex data.